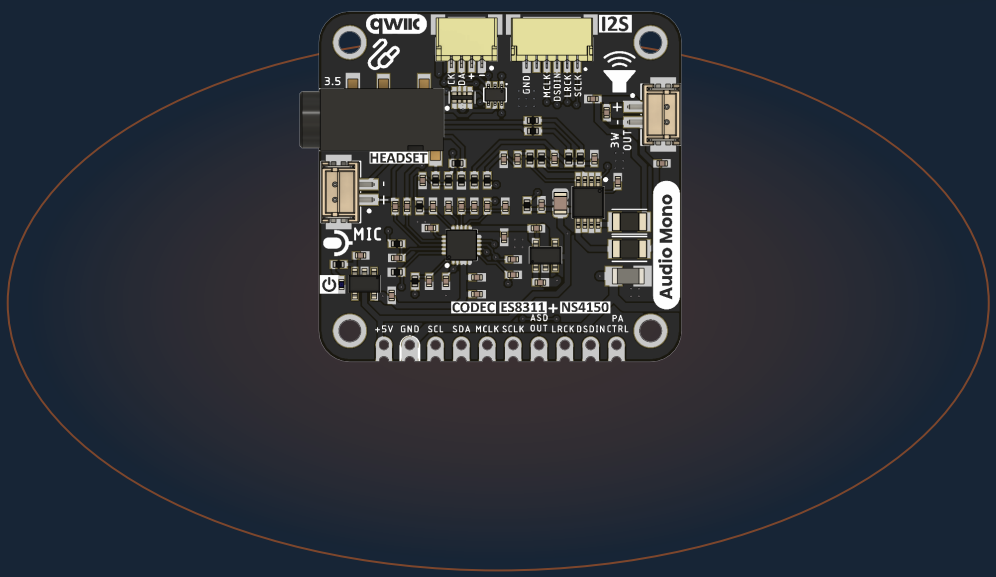
PRODUCT REFERENCE MANUAL

MEMBER OF THE UNIT DEVLAB ECOSYSTEM

UNIT I2S ES8311 Codec with Amplifier Module

MPN: UE0097

Version 0.1.0



Mono audio codec module with ES8311, NS4150B speaker amplifier, analog microphone input, 3.5 mm jack, and I2S and I2C interfaces.

ES8311 | MONO AUDIO | NS4150B | I2S · I2C · QWIIC

DOCUMENT MAP

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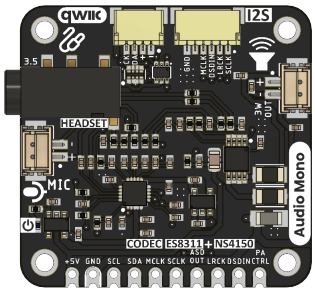
- 5 Board Operation
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Description

The UNIT I2S ES8311 Codec with Amplifier Module (UE0097) is a mono audio interface for a host controller. The ES8311 converts between analog audio and digital I2S/PCM audio. The board adds an NS4150B speaker amplifier, an analog microphone connector, a 3.5 mm audio jack, a two-pin speaker connector, an I2S header, and an I2C/QWIIC-style connector. An external host supplies clocks, audio data, control, and power.



Top view of the audio module

Applications

- Voice recording and playback in embedded systems
- Intercoms, monitoring equipment, and voice interfaces
- Audio input and output experiments with I2S-capable controllers
- Prototyping with an external analog microphone and mono speaker

Application path	Host interface	Module audio connection
Microphone capture	I2C setup, I2S clocks, ASDOUT	J2 analog microphone input
Speaker playback	I2C setup, I2S clocks, DSDIN1, PA_CTRL	J1 differential speaker output
3.5 mm audio playback	I2C setup, I2S clocks, DSDIN1	J3 analog output path
Bidirectional voice	I2C setup, I2S input and output	J2 input with J1 or J3 output

These are potential uses of the documented interfaces, not claims of application-level validation.

Hardware Features

- ES8311 mono codec with differential microphone input and I2S/PCM port
- NS4150B amplifier connected to the ES8311 differential output
- Separate speaker, microphone, 3.5 mm jack, 10-position header, and four-position QWIIC-style connectors
- Two AP2112K-3.3 regulators, power indicator, and amplifier control signal
- Four mounting holes and castellated 2.54 mm header positions

1 The Board

1.1 Accessories

Accessory	Purpose	Verified specification
I2S-capable host controller	Supplies digital audio and clocks	
Power source	Supplies VCC and ground	
Analog microphone	Connects to J2	
Mono speaker	Connects to J1	
3.5 mm plug or headset	Connects to J3	
Four-position cable	Connects to J6	

Connector mating part numbers, speaker impedance and power rating, microphone requirements, and cable orientation are not defined in the supplied board documentation.

1.2 Board Identification

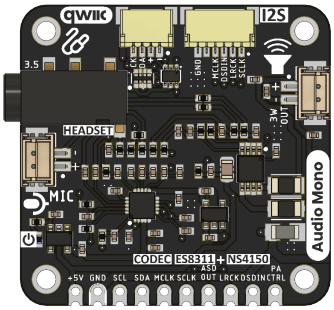
Item	Value
Product	UNIT I2S ES8311 Codec with Amplifier Module
SKU / MPN	UE0097
Product family	UNIT DevLab ecosystem
Product type	Mono audio codec and amplifier module
Main component	ES8311
Amplifier	NS4150B
Schematic revision	2.0
Image file revision	V1.1.0
Product Reference	Version 0.1.0

1.3 Main Assemblies

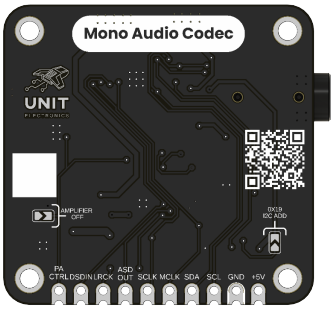
RefDes	Component	Schematic function
IC1	ES8311	Mono audio ADC, DAC, and digital audio interface
IC2	NS4150B	Differential speaker amplifier
U1, U2	AP2112K-3.3V	Separate 3.3 V regulator outputs
J1	Two-position connector	Differential speaker output
J2	Two-position connector	External microphone input

RefDes	Component	Schematic function
J3	PJ-320D	3.5 mm audio jack
J4	10-position castellated header	Power, I2C, I2S, and amplifier control
J6	Four-position connector	Power, ground, SDA, and SCL
D1	Green LED	Power indicator

1.4 Board Views



Top view of the module



Bottom view of the module

The images show connector placement and silkscreen. For electrical connections in this reference, use the revision 2.0 schematic.

1.5 Handling

Observe normal ESD precautions. Remove power before changing microphone, speaker, jack, or host wiring. Verify connector pin 1, polarity, and voltage with the actual hardware revision before applying power.

2 Ratings

The ES8311 supply and audio figures in this chapter come from its User Guide. Blank cells indicate undocumented module ratings.

2.1 Recommended Operating Conditions

Parameter	Minimum	Typical	Maximum
ES8311 AVDD	1.7 V	3.3 V	3.6 V
ES8311 PVDD	1.6 V	3.3 V	3.6 V
ES8311 DVDD	1.6 V	3.3 V	3.6 V
Module VCC input			
Module 3V3 output current			
Module speaker load			
Module operating temperature			

The board image labels the header supply as +5V. This identifies the intended nominal input marking, but the supplied documents give no validated module input range. The ES8311 supply values above do not define VCC limits because VCC also supplies both regulators and the amplifier.

2.2 Power Domains and Power Distribution

The schematic routes VCC to the NS4150B amplifier, the power LED, J4 and J6, and the inputs of U1 and U2. U1 produces the 3V3 net; U2 produces A_3V3. The ES8311 power pins and I2C address pull-up use 3V3. Its AVDD pin uses A_3V3. Analog and digital grounds meet through the shown zero-ohm coupling element.

2.3 Digital Interfaces

Interface	Board connection	Documented behavior
I2C control	J4 SDA/SCL and J6 SDA/SCL	ES8311 CE is pulled to 3V3 through R5; guide maps this to 7-bit address 0x19
Digital audio	J4 MCLK, SCLK, LRCK, ASDOUT, DSDIN1	ES8311 supports I2S, left-justified, and DSP-A/B formats
Amplifier enable	J4 PA_CTRL	Connected to NS4150B CTRL through R12
Digital microphone	ES8311 multiplexed pins	Component capability; no external digital-microphone connector is documented

The guide describes ES8311 master and slave audio modes. It does not establish which mode a particular host application must use.

2.4 Confirmed Component Values

Parameter	Value
ADC signal-to-noise ratio	100 dB
DAC signal-to-noise ratio	110 dB
ADC output sample rate	Up to 96 kHz

2.5 Application-Dependent and Uncharacterized Parameters

Parameter	Module value
Total operating current	
Amplifier output power and distortion	
Validated speaker impedance	
Microphone bias voltage and supported microphone types	
Input gain and full-scale input at J2/J3	
Module audio signal-to-noise ratio	
End-to-end latency and noise	
ESD, EMC, and environmental limits	

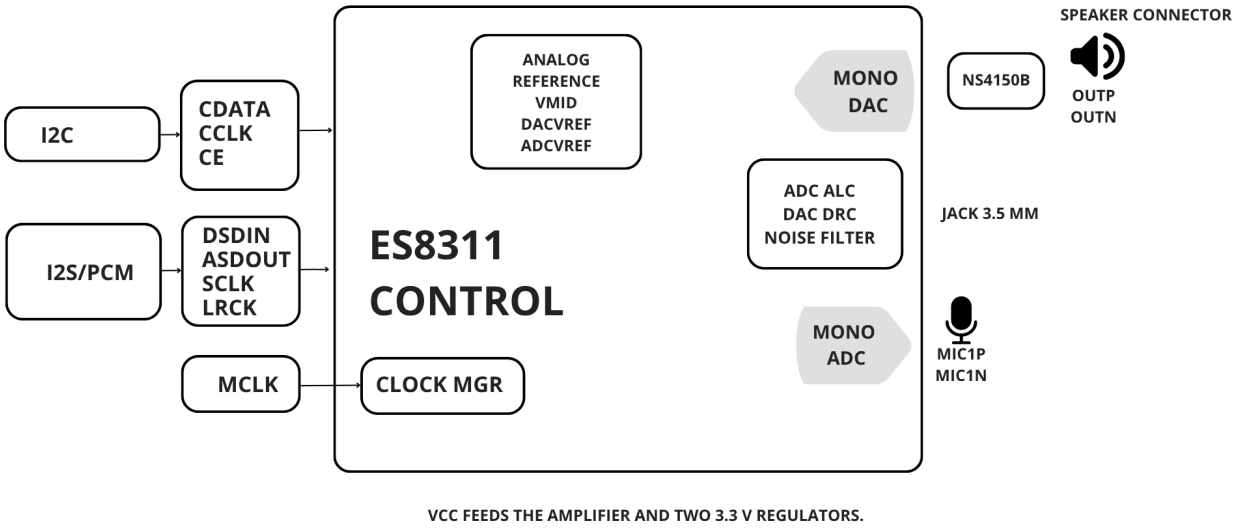
2.6 Electrical Precautions

1. Keep host ground common with the module and verify the VCC rail before connection. The +5V label does not document an allowable voltage range.
2. J6 carries VCC in the schematic. Do not assume its supply is the 3.3 V normally associated with QWIIC accessories; check the connected device.
3. Do not derive a safe speaker load or amplifier output power from the ES8311 headphone-driver specification. The speaker is driven by NS4150B.
4. Do not connect a speaker output to a host logic or line-level input.

3 Functional Overview

The external host supplies power and I2C control and exchanges serial audio with the ES8311. The direction of the serial audio clocks depends on the codec mode. The ES8311 handles mono recording and playback. Its differential analog output feeds both a jack path and the NS4150B speaker amplifier in the schematic.

3.1 Block Diagram



Functional block diagram of the ES8311 codec, microphone input, jack path, and NS4150B speaker amplifier

3.2 Board Topology

The top image shows the QWIIC-style and I2S connectors along one edge, the speaker connector, microphone connector, 3.5 mm jack, and a ten-position castellated header. The opposite image shows the amplifier-off and I2C-address markings.

3.3 Audio Codec

IC1 is the ES8311. Its recording path includes a differential analog input, microphone preamplifier, ADC, programmable gain, and automatic level control. The playback path includes a mono DAC and differential analog outputs. Configuration uses the I2C control interface. See the ES8311 User Guide for register-level programming and valid clock configurations.

3.4 Digital Audio Architecture

The serial audio port supports I2S, left-justified, and DSP-A/B formats. In each format, the codec sends the most significant data bit first. DSDIN1 receives playback samples for the DAC; ASDOUT sends captured ADC samples to the host. These data directions do not change with the clock mode. SCLK times the data bits, while LRCK identifies the audio frame alignment. The host and codec must use the same data format and word length.

Codec mode	SCLK and LRCK at J4	Clock source at these pins
Slave (reset default)	Inputs	Host
Master	Outputs	ES8311

MCLK is an input in either mode. The codec can select MCLK or SCLK as its internal master-clock source through register settings. The ES8311 User Guide, sections 7 and 8, defines the format timing and clock-divider options.

3.5 Speaker Amplifier

IC2 is NS4150B. ES8311 OUTP and OUTN are AC-coupled into its differential input. Amplifier outputs PA_OUT+ and PA_OUT− reach J1 through zero-ohm links. PA_CTRL is exposed to the host and connects to the amplifier CTRL input.

3.6 Analog Microphone Input

J2 exposes the microphone input circuit for ES8311 MIC1P and MIC1N. The schematic includes resistors and capacitors around the input and bias network.

3.7 Status Indicator

D1 is a green power indicator connected from VCC through R9 to ground. It indicates presence of the module supply; it does not report codec or amplifier configuration.

3.8 AP2112K Power Management System

U1 and U2 are AP2112K-3.3V regulators. U1 creates 3V3 and U2 creates A_3V3 from VCC.

3.9 Power Tree

Source	Destination	Schematic connection
VCC	IC2, D1, J4, J6, U1, U2	Direct module supply net
U1 / 3V3	IC1 digital supply and I2C pull-up	Regulated net
U2 / A_3V3	IC1 AVDD	Analog regulated net
AGND and GND	R8	Zero-ohm ground coupling

3.10 I2C Control

I2C reads and writes codec registers; audio samples travel on the separate serial audio port. J4 pin 3 (SCL) and J6 pin 4 share the clock net, which reaches IC1 CCLK through zero-ohm R7. J4 pin 4 (SDA) and J6 pin 3 share the bidirectional data net, which reaches IC1 CDATA through zero-ohm R4.

R5 pulls CE to 3V3, selecting the 7-bit address 0x19 in the ES8311 User Guide. The guide specifies an I2C transfer rate below 400 kbit/s and recommends 1–4.7 kΩ pull-ups for CCLK and CDATA. The available schematic page does not show dedicated SCL/SDA pull-ups. Verify the pull-ups and their voltage on the assembled bus before connecting the host or another I2C device.

3.11 Headphone / Jack Path

The ES8311 differential outputs also reach J3, a PJ-320D 3.5 mm jack, through the schematic's coupling and link components. MIC1P and ground are also present at the jack circuit.

3.12 I2C and QWIIC Expansion

J6 extends the same I2C bus as J4; it is not a separate controller or an I2S connection. Devices attached to J6 share the bus pull-ups and address space. J6 also carries the module VCC input net, not the regulated 3V3 net. Check the attached device's supply and logic compatibility before connecting it.

3.13 Combined System Operation

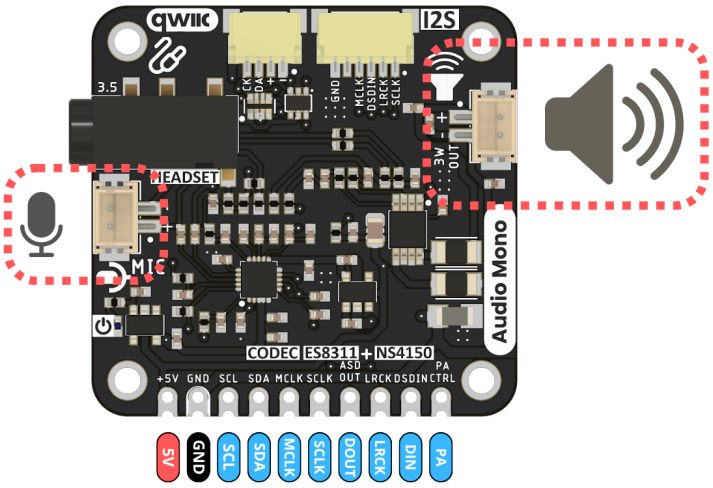
A host configures the ES8311 by I2C, supplies suitable clocks, then transfers digital audio over the serial port. The host separately controls PA_CTRL when using the speaker amplifier.

4 Connectors & Pinouts

This chapter transcribes the revision 2.0 schematic. Pin numbering is electrical numbering in that schematic. Before wiring physical connectors, confirm pin-1 orientation on the actual board revision.

4.1 J4 Host Header Pinout

J4 is the ten-position castellated host header.



Directions are relative to the module.

J4 pin	Signal	Direction	Function
1	VCC	Input	Module supply (+5V marking)
2	GND	Return	Ground reference
3	SCL	Input	I2C clock from host
4	SDA	Bidirectional	I2C register data
5	MCLK	Input	Codec master clock
6	SCLK	Input / output	Serial audio bit clock
7	ASDOUT	Output	ADC samples to host
8	LRCK	Input / output	Audio frame clock
9	DSDIN1	Input	DAC samples from host
10	PA_CTRL	Input	NS4150B amplifier control

SCLK and LRCK are inputs in slave mode and outputs in master mode.

The illustration abbreviates ASDOUT as DOUT, DSDIN1 as DIN, and PA_CTRL as PA. The board silkscreen abbreviates ASDOUT as OUT. These are digital audio and control signals, not the speaker output.

4.2 Host Pinout Compatibility

Host requirement	Module connection	Validated host mapping
I2C SCL / SDA	J4 pins 3 / 4	
I2S MCLK / SCLK / LRCK	J4 pins 5 / 6 / 8	
I2S ADC receive	J4 pin 7	
I2S DAC transmit	J4 pin 9	
Amplifier control	J4 pin 10	
Supply / return	J4 pins 1 / 2	

The module does not implement an Arduino Nano pinout. Host GPIO numbers, voltage compatibility, and required firmware depend on the chosen controller.

4.3 QWIIC Connector

J6 pin	Schematic net
1	VCC
2	AGND
3	SDA
4	SCL

The board calls J6 QWIIC, while the schematic connects its supply contact to VCC. Connector pitch, mating part, and safe voltage for attached devices are not established by the supplied documentation.

4.4 Microphone Connector

J2 is a two-position connector for the external analog microphone circuit. The schematic shows pin 1 on the MIC1P path and pin 2 on the MIC1N path. Connector polarity and microphone electrical requirements must be checked against the actual assembly.

4.5 Speaker Connections

J1 is the two-position speaker connector: pin 1 is OUT+ and pin 2 is OUT−. These are the differential outputs of the NS4150B amplifier. Neither contact is a ground return.

4.6 3.5 mm Audio Jack

J3 is a PJ-320D jack connected to the codec output path, MIC1P circuit, and analog ground. The supplied schematic does not define a tested headset wiring standard or a user-facing plug-contact pinout.

4.7 Power and Amplifier Control

VCC enters through J4 and is also present at J6. There is no USB, BOOT, or reset circuit on this module. J4 pin 10 carries PA_CTRL; the schematic connects it to IC2 CTRL through R12. The required logic thresholds and safe power-up sequence are not specified at module level.

4.8 Test Pads and Assembly Options

The images show solder options labelled amplifier off and I2C address. The revision 2.0 schematic shows a PA_CTRL-to-ground option and an ES8311 CE pull-up. The default assembled configuration and rework instructions for those options are not documented. Confirm the populated board before changing either option.

5 Board Operation

The module is a peripheral; setup is performed on an external host. The ES8311 User Guide provides component-level register and clock information. The repository currently contains no verified module-specific firmware example.

5.1 Getting Started with a Host Controller

Choose a host with an I2C controller and serial audio interface. Connect supply, ground, SCL, SDA, serial audio data, and PA_CTRL to the J4 functions in Chapter 4. Connect MCLK when selected as the codec's internal clock source, and set the host's SCLK and LRCK direction to match the mode in Section 3.4. Confirm power and logic compatibility before energizing the board. The correct host pin numbers and codec driver depend on the host platform.

5.2 First Power-up and Inspection

Inspect connector orientation and the actual revision. Apply the supply intended for that assembly using a current-limited source. The power LED should indicate VCC presence. Verify 3V3 and A_3V3 before connecting a speaker or microphone. The supplied documents contain no module current limit or production acceptance test.

5.3 Control and Clock Workflow

Use I2C to detect the codec at the 7-bit address 0x19 selected by the revision 2.0 schematic's CE pull-up. Configure the serial audio format, word length, clock source, and clock dividers using the ES8311 User Guide. After reset the codec defaults to slave mode, so the host drives SCLK and LRCK. If master mode is selected, the codec drives those lines; the host must not drive them at the same time. MCLK remains an input when selected as the clock source. Enable the record and playback paths required by the application. A board-tested register sequence has not been supplied.

5.4 Recommended Bring-up Sequence

Step	Check	Expected observation
1	Power	VCC, 3V3, and A_3V3 present
2	Control	ES8311 responds at the configured I2C address
3	Clocks	MCLK, SCLK, and LRCK follow the selected mode
4	Playback	DSDIN1 carries audio data
5	Record	ASDOUT carries audio data
6	Amplifier	PA_CTRL selects the desired amplifier state
7	External audio	J1, J2, and J3 work with verified loads

Measured pass/fail ranges for these checks are not published.

5.5 Firmware Example Collection

Platform	Verified example
Arduino C++	
MicroPython	
Other hosts	

No module-specific example has been validated in this repository.

5.6 Audio Buffer Planning

Allocate host-side buffers for the selected sample format, clock rate, and latency target. The module itself has no application RAM or storage. Buffer size, DMA strategy, and supported host configurations depend on the host system and are not specified here.

5.7 Recording Operation

J2 supplies the ES8311 analog microphone path. Configure input selection, gain, filtering, and ADC output format using the User Guide. Confirm the microphone's electrical needs separately. There is no board-validated microphone model or sensitivity figure in the supplied sources.

5.8 Playback and Recording Coexistence

The ES8311 supports simultaneous ADC and DAC paths at component level. The module routes playback toward the jack and amplifier. A host application must coordinate clocks, serial audio slots, buffer flow, and PA_CTRL. No combined-system performance or tested configuration is documented.

5.9 Troubleshooting Guide

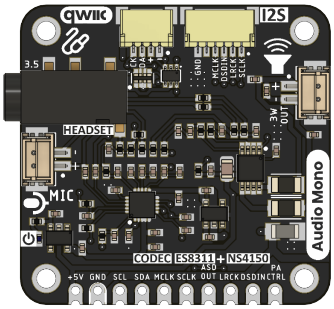
Symptom	Check
No I2C response	VCC, ground, 3V3, J4 pin 3 SCL / pin 4 SDA, bus pull-ups, CE population, 7-bit address
No digital audio	Clock source, SCLK/LRCK direction, matching format and word length, DSDIN1/ASDOUT routing
No speaker sound	PA_CTRL, speaker on J1, amplifier VCC, playback path
No microphone signal	J2 wiring, input bias, ADC path, gain and routing
Unexpected noise	Supply and ground arrangement, clock wiring, analog connections

These are diagnostic checks, not validated fault signatures.

6 Mechanical Information

The available top and bottom board images show a compact module with four mounting holes, a 10-position castellated header, a 3.5 mm jack, and audio connectors. They are tagged V1.1.0 and are illustrative for the revision 2.0 electrical reference.

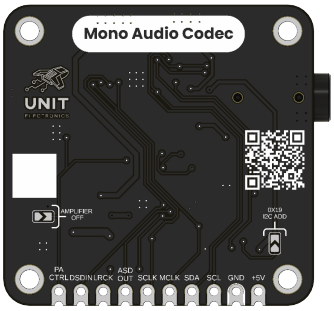
6.1 Component-side Envelope



Component side

Allow clearance for the 3.5 mm plug, microphone and speaker plugs, the QWIIC-style and I2S connectors, and host wiring.

6.2 Bottom-side Envelope



Bottom side

Maintain clearance around solder pads and mounting holes. The available image does not establish a controlled hole pattern or board thickness.

6.3 Mechanical Data Not Specified

Dimension	Value
Board length	
Board width	
PCB thickness	

Dimension	Value
Mounting-hole diameter	
Mounting-hole center coordinates	
Maximum component height, top	
Maximum component height, bottom	
Connector keep-outs	

Use a controlled mechanical drawing or physical measurement before designing an enclosure or carrier. No such drawing is supplied in the available package.

7 Company Information

Item	Information
Company	UNIT Electronics
Website	https://uelectronics.com/
Product-specific support contact	
Product-specific warranty terms	

The product repository and source documents are listed in Chapter 8.

8 Reference Documentation

Reference	Location	Use in this Product Reference
Board schematic, revision 2.0	hardware/resources/ unit_sch_v_2_0_0_i2s_es8311_codec_with_amplifier_module.pdf	Board connections and component identifiers
ES8311 User Guide, Rev 1.11, September 2018	hardware/resources/es8311_user_guide.pdf	Codec capabilities, supply conditions, I2C address, and audio interface
Top board image, V1.1.0	hardware/img/ unit_top_v_1_1_0_i2s_es8311_codec_with_amplifier_module.png	Visual identification and labels
Bottom board image, V1.1.0	hardware/img/ unit_bottom_v_1_1_0_i2s_es8311_codec_with_amplifier_module.png	Visual identification and labels
Module-specific amplifier datasheet		
Controlled mechanical drawing		
Verified firmware example		

The local PDFs are source documents, not generated by this Product Reference build. The ES8311 guide describes the codec component; it does not certify the electrical or acoustic performance of the assembled module.

9 Appendix

9.1 Schematic

The source drawing is Revision 2.0 schematic. Its title block states sheet 1 of 2, while the supplied PDF contains one page. A second page has not been supplied.

9.1.1 Codec, Power, Amplifier, and Connectors

The available PDF page includes IC1 ES8311, IC2 NS4150B, U1/U2 regulators, J1/J2/J3/J4/J6, microphone bias, and ground coupling.

9.1.2 Additional Schematic Sheet

Field	Value
Additional sheet file	
Additional circuits	

9.1.3 Controlled Mechanical Drawing

Field	Value
Drawing file	
Drawing revision	

9.2 Technical References

1. The board schematic revision 2.0 defines the circuit and connector signal names.
2. The ES8311 User Guide Rev 1.11 defines codec component characteristics and register-level interface behavior.
3. The V1.1.0 top and bottom images identify visible parts and labels but do not validate the revision 2.0 electrical implementation.

9.3 Values Not Specified by the Technical Documentation

Missing board-level item	Value
Validated VCC input range	
Module current and power budget	
Speaker load and amplifier power rating	
Microphone electrical compatibility	
3.5 mm plug-contact standard	
Connector mating part numbers	
Overall dimensions and mounting coordinates	
Environmental limits and qualification	

Missing board-level item	Value
Tested codec initialization and firmware	
Rev 2.0 assembly-to-image correspondence	

9.4 Document Control

Field	Value
Product	UNIT I2S ES8311 Codec with Amplifier Module
SKU / MPN	UE0097
Schematic revision	2.0
Board image file revision	V1.1.0
Product Reference	Version 0.1.0
Publication date	2026-10-19
Firmware revision	

9.5 Source Inconsistencies

- The board image filenames contain V1.1.0; the schematic title block identifies revision 2.0. The available sources do not reconcile them.
- The schematic title block identifies sheet 1 of 2, but the supplied PDF has one page.
- The images call J6 QWIIC and mark the board supply +5V. The schematic connects J6 supply to VCC, not to the regulated 3V3 net.
- The ES8311 guide cover displays Rev 1.00 while its page footers display Rev 1.11. Component facts in this document are cited to the supplied file.

9.6 Hardware Errata

Erratum	Status	Resolution

No verified hardware erratum is included in the supplied sources.